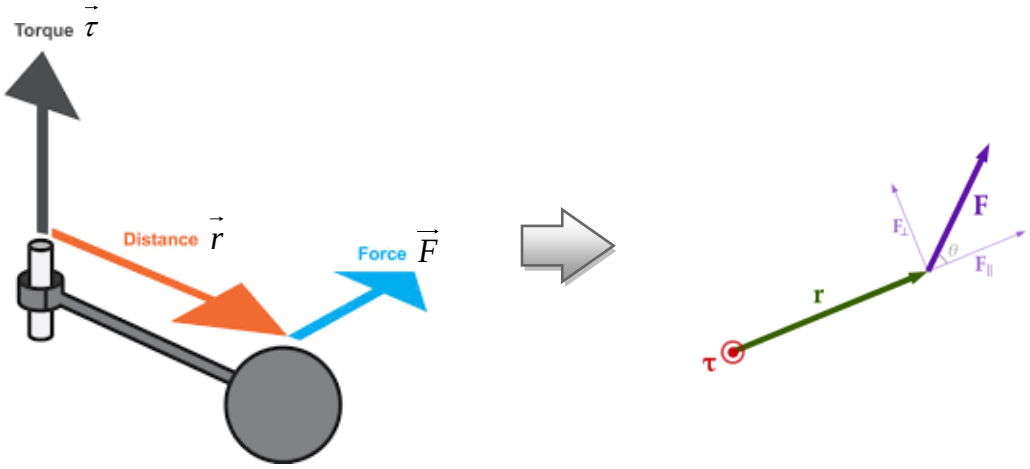
Vector Product (Cross Product)

Vectors are widely used in physics, especially in mechanics. For example, the torque (or moment) $\vec{\tau}$ produced by constant turning force $\vec{F}$ applied with a length of the level arm $\vec{r}$ is given by



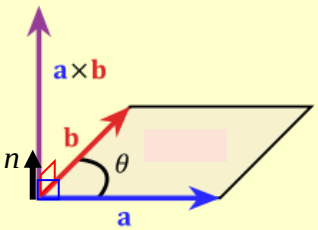
$$\vec{\tau} = |\vec{r}||\vec{F}|\sin\theta$$ (note that $\vec{\tau}$ is a vector)

Since the above operation is applied frequently, in order to have a better representation, **vector product (向量積) / cross product (叉積)** is introduced, i.e.

Vector Product (Cross Product)

The **vector product** (also refer as **cross product**) of two vectors $\vec{a}$ and $\vec{b}$, denoted as $\vec{a}\times\vec{b}$, is defined as:

$$\vec{a}\times\vec{b} = |\vec{a}||\vec{b}|\sin\theta\, n$$



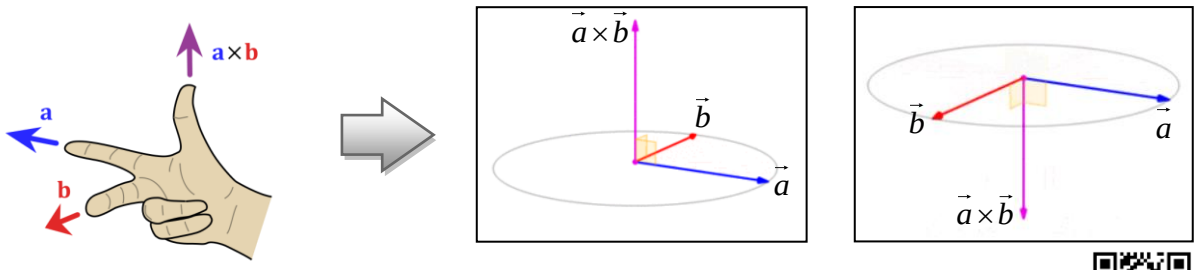
where θ is the angle between $\vec{a}$ and $\vec{b}$, and $0° \le \theta \le 180°$;

n is the unit vector perpendicular to the plane containing $\vec{a}$ and $\vec{b}$

Not for sale

Definition of Vector Product

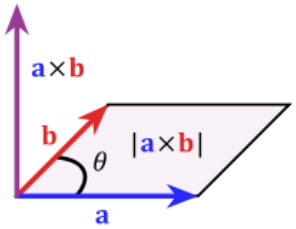
For any two free vectors $\vec{a}$ and $\vec{b}$, we can determine the direction of $\vec{a}\times\vec{b}$ by using the "right-hand rule":



Reference: https://en.wikipedia.org/wiki/Cross_product

Recall that $\vec{a}\times\vec{b} = |\vec{a}||\vec{b}|\sin\theta\, n$, therefore the magnitude of $\vec{a}\times\vec{b}$ (i.e. $|\vec{a}\times\vec{b}|$) can be determined as follows:

$$|\vec{a}\times\vec{b}| = |\vec{a}||\vec{b}|\sin\theta$$



And note that

- $|\vec{a}\times\vec{b}|$ is the **area of the parallelogram** formed by vectors $\vec{a}$ and $\vec{b}$.
- $|\vec{a}\times\vec{b}|$ is minimum (i.e., zero) when $\theta = 0°$ or $\theta = 180°$ (i.e. $\vec{a}\,//\,\vec{b}$).
- $|\vec{a}\times\vec{b}|$ is maximum when $\theta = 90°$.

For any **vectors** **a** , **b**, **c** and **constants** *k*, we have the following properties:

(a) $\mathbf{a}\times\mathbf{a} = \mathbf{0}$

(b) $\mathbf{a}\times\mathbf{b} = -\mathbf{b}\times\mathbf{a}$

(c) $\mathbf{a}\times\mathbf{b} = \mathbf{0}$ iff **a** and **b** are parallel vectors.

(d) $\mathbf{a}\times(\mathbf{b}+\mathbf{c}) = \mathbf{a}\times\mathbf{b} + \mathbf{a}\times\mathbf{c}$

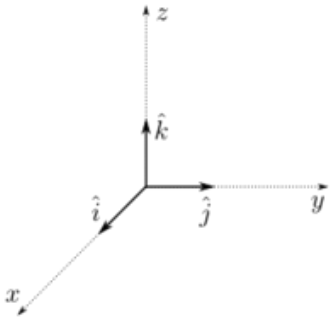
(e) $(\mathbf{a}+\mathbf{b})\times\mathbf{c} = \mathbf{a}\times\mathbf{c} + \mathbf{b}\times\mathbf{c}$

(f) $k(\mathbf{a}\times\mathbf{b}) = (k\mathbf{a})\times\mathbf{b} = \mathbf{a}\times(k\mathbf{b})$

Not for sale

Vector Product for Base Unit Vectors

In the coordinates system, there are three **base unit vectors**:



- $\vec{i}$ represents the unit vector in the **+ve direction of *x*-axis**
- $\vec{j}$ represents the unit vector in the **+ve direction of *y*-axis**
- $\vec{k}$ represents the unit vector in the **+ve direction of *z*-axis**

And there are some basic properties of base unit vectors:

(i) $\vec{i}\times\vec{i} = \vec{j}\times\vec{j} = \vec{k}\times\vec{k} = \vec{0}$

(ii) $\vec{i}\times\vec{j} = \vec{k}$; $\vec{j}\times\vec{k} = \vec{i}$; $\vec{k}\times\vec{i} = \vec{j}$

(iii) $\vec{j}\times\vec{i} = -\vec{k}$; $\vec{k}\times\vec{j} = -\vec{i}$; $\vec{i}\times\vec{k} = -\vec{j}$

Therefore if $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$ and $\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$ are vectors in R^3, then the vector product $\vec{a}\times\vec{b}$ can be expressed by

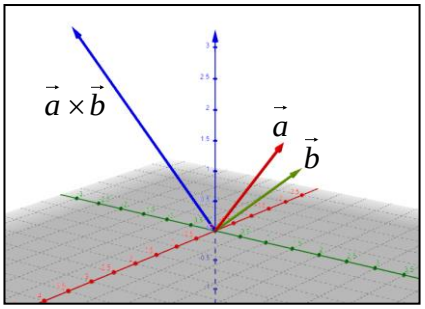
$$\vec{a}\times\vec{b} = (a_2b_3 - a_3b_2)\vec{i} + (a_3b_1 - a_1b_3)\vec{j} + (a_1b_2 - a_2b_1)\vec{k}$$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

For example, if $\vec{a} = \vec{i} + 2\vec{j} + 2\vec{k}$ and $\vec{b} = -\vec{i} + \vec{j} + \vec{k}$,

then $\vec{a}\times\vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & 2 \\ -1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} 2 & 2 \\ 1 & 1 \end{vmatrix}\vec{i} - \begin{vmatrix} 1 & 2 \\ -1 & 1 \end{vmatrix}\vec{j} + \begin{vmatrix} 1 & 2 \\ -1 & 1 \end{vmatrix}\vec{k}$

$= -3\vec{j} + 3\vec{k}$



Not for sale

Vector Product for Unit Base Vectors



Let's Try:

1. It is given that $\mathbf{a} = 2\mathbf{i} + 3\mathbf{j} - \mathbf{k}$ and $\mathbf{b} = \mathbf{i} - 2\mathbf{j} + 2\mathbf{k}$. Find $\mathbf{a} \times \mathbf{b}$.

2. It is given that $\mathbf{a} = 6\mathbf{i} + 3\mathbf{k}$ and $\mathbf{b} = \mathbf{i} + \mathbf{j} - 2\mathbf{k}$.
   (a) Find $\mathbf{a} \cdot \mathbf{b}$.
   (b) Find $\mathbf{a} \times \mathbf{b}$.

Not for sale

Application: Areas of Δ in $\mathbb{R}^3$

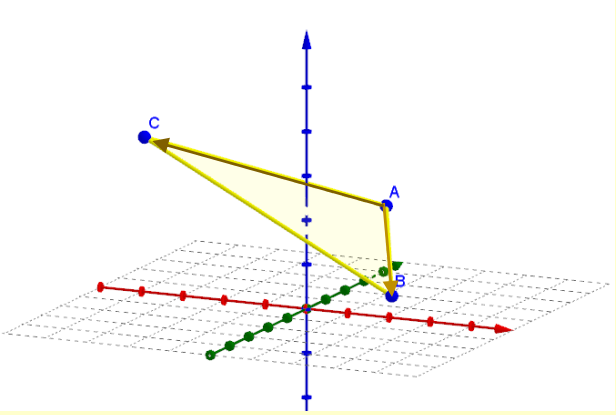
For any three points A, B and C in $\mathbf{R}^3$,

$$\text{Area of } \Delta ABC$$

$$= \frac{1}{2}|\overrightarrow{AB}||\overrightarrow{AC}|\sin\angle BAC$$

$$= \frac{1}{2}|\overrightarrow{AB}\times\overrightarrow{AC}|$$

(or $\frac{1}{2}|\overrightarrow{BA}\times\overrightarrow{BC}|$ or $\frac{1}{2}|\overrightarrow{CA}\times\overrightarrow{CB}|$)



e.g. The coordinates of A, B and C are (1, 1, 3), (3, –1, 5) and (–1, 0, 1) respectively. Find the area of $\triangle ABC$.

$$\overrightarrow{AB} = (3-1)\vec{i} + [(-1)-1]\vec{j} + (5-3)\vec{k}$$
$$= 2\vec{i} - 2\vec{j} + 2\vec{k}$$
$$\overrightarrow{AC} = [(-1)-1]\vec{i} + (0-1)\vec{j} + [(-1)-3]\vec{k}$$
$$= -2\vec{i} - \vec{j} - 4\vec{k}$$
$$\overrightarrow{AB}\times\overrightarrow{AC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -2 & 2 \\ -2 & -1 & -4 \end{vmatrix} = 10\vec{i} + 4\vec{j} + 6\vec{k}$$

The "| |" here means determinant

$$\therefore \quad \text{Area of } \Delta ABC = \frac{1}{2}|\overrightarrow{AB}\times\overrightarrow{AC}|$$

The "| |" here means magnitude of vector

$$= \frac{1}{2}\sqrt{(10)^2 + (4)^2 + (6)^2}$$
$$= \sqrt{38}$$

Not for sale

Application: Distance between a point and a line in R³

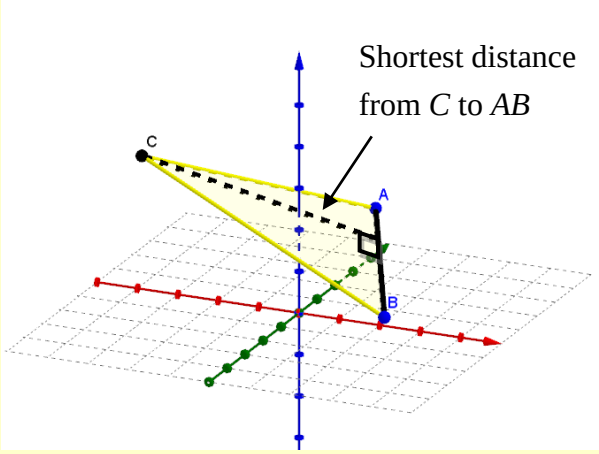
For any three points A, B and C in $\mathbf{R}^3$,

Distance (d) from C to AB = Altitude from C to AB in ΔABC

$\because$ Area of $\Delta ABC = \dfrac{|\overrightarrow{AB}| \cdot d}{2}$

$\therefore \quad \dfrac{1}{2} |\overrightarrow{AB} \times \overrightarrow{AC}| = \dfrac{|\overrightarrow{AB}| \cdot d}{2}$

$$d = \frac{|\overrightarrow{AB} \times \overrightarrow{AC}|}{|\overrightarrow{AB}|}$$



e.g. The coordinates of A, B and C are (1, 1, 3), (3, –1, 5) and (–1, 0, 1) respectively. Find the shortest distance from C to AB

From the work on the pervious page,

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -2 & 2 \\ -2 & -1 & -4 \end{vmatrix} = 10\vec{i} + 4\vec{j} + 6\vec{k}$$

$$|\overrightarrow{AB} \times \overrightarrow{AC}| = \sqrt{(10)^2 + (4)^2 + (6)^2} = 2\sqrt{38}$$

$$|\overrightarrow{AB}| = \sqrt{(2)^2 + (-2)^2 + (2)^2} = 2\sqrt{3}$$

$\therefore$ shortest distance from C to $AB = \dfrac{2\sqrt{38}}{2\sqrt{3}} = \dfrac{\sqrt{114}}{3}$

Not for sale

Applications of Vector Product



Let's Try:

1. Let $A(3, -1, 4)$, $B(6, -4, 4)$ and $C(2, 1, 2)$ be three points in $\mathbf{R}^3$.

   (a) Express $\overrightarrow{AB}$ and $\overrightarrow{AC}$ in terms of **i**, **j** and **k**.

   (b) Find the area of $\triangle ABC$.

   (c) Hence, or otherwise, find the shortest distance from B to AC.

Not for sale

Application: Volume of parallelepiped / tetrahedron

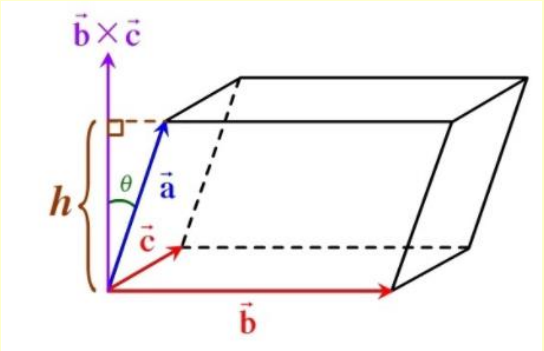
Volume of parallelepiped (平行六面體) *[Out of Syllabus]*

= base area × height

$= (|\vec{b}\times\vec{c}|)(|\vec{a}|\cos\theta)$

$= \left|\vec{a}\cdot(\vec{b}\times\vec{c})\right|$

(or $\left|\vec{b}\cdot(\vec{c}\times\vec{a})\right|$ or $\left|\vec{c}\cdot(\vec{a}\times\vec{b})\right|$)



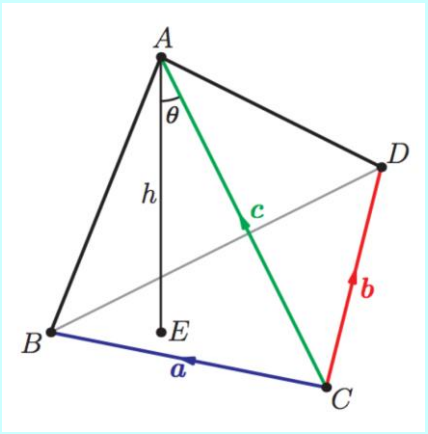
Volume of tetrahedron (四面體)

$= \frac{1}{3}\times$ base area × height

$= \frac{1}{3}\cdot\frac{|\vec{a}\times\vec{b}|}{2}(|\vec{c}|\cos\theta)$

$= \frac{1}{6}\left|\vec{c}\cdot(\vec{a}\times\vec{b})\right|$

(or $\frac{1}{6}\left|\vec{b}\cdot(\vec{c}\times\vec{a})\right|$ or $\frac{1}{6}\left|\vec{a}\cdot(\vec{b}\times\vec{c})\right|$)



Note:

1. $\vec{a}\cdot(\vec{b}\times\vec{c})$ is called a **scalar triple product (純量三重積)** of the three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$.
2. $\vec{a}\cdot(\vec{b}\times\vec{c})=0$ iff $\vec{a}$, $\vec{b}$ and $\vec{c}$ are **coplanar (共面)**.

Applications of Vector Product

[M2 2012 Q.7]

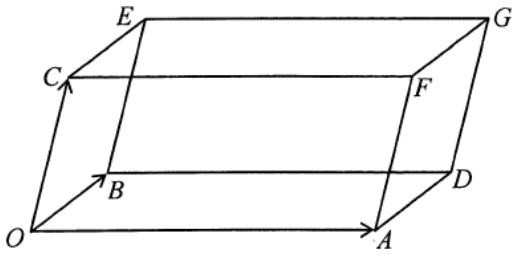


Figure 3

Figure 3 shows a parallelepiped *OADBECFG*. Let $\overrightarrow{OA} = 6\mathbf{i} + 2\mathbf{j} - \mathbf{k}$, $\overrightarrow{OB} = 2\mathbf{i} + \mathbf{j}$ and $\overrightarrow{OC} = 5\mathbf{i} - \mathbf{j} + 2\mathbf{k}$.

(a) Find the area of the parallelogram *OADB*.

(b) Find the distance between point *C* and the plane *OADB*. ***(5 marks)***

Not for sale

Application: Distance between a point and a plane

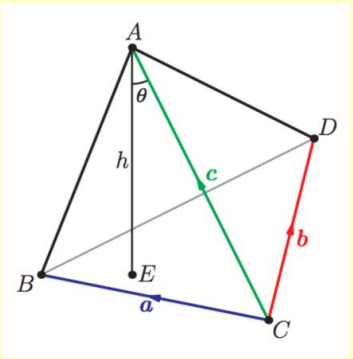
For any point A and plane (e.g. triangular plane BCD) C in $\mathbf{R}^3$,

Let h be the distance from A to ΔBCD

$\because$ Volume of tetrahedron $ABCD = \frac{1}{3}$(Area of ΔBCD) h

$$\therefore \quad \frac{1}{6}\left|\vec{c}\cdot(\vec{a}\times\vec{b})\right| = \frac{1}{3}\frac{|\vec{a}\times\vec{b}|}{2}h$$

$$h = \frac{|\vec{c}\cdot(\vec{a}\times\vec{b})|}{|\vec{a}\times\vec{b}|}$$



Note: In the figure, the vector $\overrightarrow{EA}$ can be expressed as either

$$\frac{|\vec{c}\cdot(\vec{a}\times\vec{b})|}{|\vec{a}\times\vec{b}||\vec{a}\times\vec{b}|}(\vec{a}\times\vec{b}) \quad \text{OR} \quad -\frac{|\vec{c}\cdot(\vec{a}\times\vec{b})|}{|\vec{a}\times\vec{b}||\vec{a}\times\vec{b}|}(\vec{a}\times\vec{b})$$

e.g. $O(0, 0, 0)$, $A(3, -1, 4)$, $B(6, -4, 4)$ and $D(2, 1, 2)$ are four points in $\mathbf{R}^3$. Find the distance between A and ΔBCD.

$$\overrightarrow{OB}\times\overrightarrow{OD} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 6 & -4 & 4 \\ 2 & 1 & 1 \end{vmatrix} = -8\vec{i}+2\vec{j}+14\vec{k}$$

$$\overrightarrow{OA}\cdot(\overrightarrow{OB}\times\overrightarrow{OD}) = (3)(-8)+(-1)(2)+(4)(14) = 30$$

$$|\overrightarrow{OB}\times\overrightarrow{OD}| = \sqrt{(-8)^2+(2)^2+(14)^2} = 2\sqrt{66}$$

$\therefore$ Distance between A and ΔBCD $= \dfrac{|\overrightarrow{OA}\cdot(\overrightarrow{OB}\times\overrightarrow{OD})|}{|\overrightarrow{OB}\times\overrightarrow{OD}|}$

$$= \frac{30}{2\sqrt{66}} = \frac{5\sqrt{66}}{22}$$

Not for sale

Applications of Vector Product

[M2 2023 Q.10 (c) Modified]

Let O be the origin. The position vectors of P and Q are $-2\mathbf{i} - \mathbf{k}$ and $2\mathbf{i} - \mathbf{j} + \mathbf{k}$ respectively. Let Π be the plane which contains O, P and Q.

(i) Find a non-zero vector which is perpendicular to Π.

(ii) Denote the projection of point $A(-6, -22, 2)$ on Π by B. Find $\overrightarrow{OB}$.

(4 marks)

Not for sale

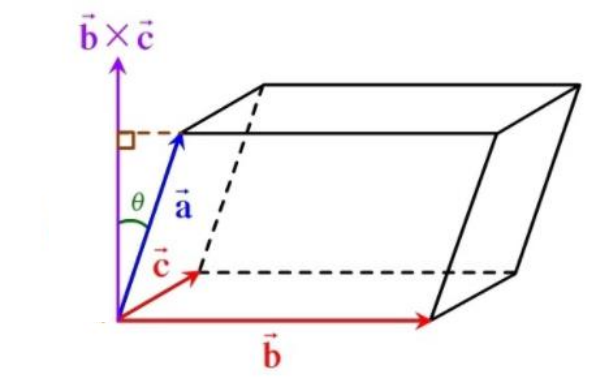
Application: Angle between a line and a plane

Let ϕ be the angle between $\vec{a}$ and the plane consisting of $\vec{b}$ and $\vec{c}$,
θ be the angle between $\vec{a}$ and $\vec{b}$ and $\vec{c}$ (note that $\theta + \phi = 90°$)

Then

$$\cos\theta = \frac{|\vec{a}\cdot(\vec{b}\times\vec{c})|}{|\vec{a}||\vec{b}\times\vec{c}|}$$

$$\therefore \quad \sin\phi = \frac{|\vec{a}\cdot(\vec{b}\times\vec{c})|}{|\vec{a}||\vec{b}\times\vec{c}|}$$



(OR you may consider $\sin\phi = \frac{h}{|\vec{a}|}$, where $h = \frac{|\vec{a}\cdot(\vec{b}\times\vec{c})|}{|\vec{b}\times\vec{c}|}$.)

e.g. Suppose $\vec{a} = 2\vec{i}+3\vec{j}-3\vec{k}$, $\vec{b} = 3\vec{i}-2\vec{j}+\vec{k}$ and $\vec{c} = \vec{i}+4\vec{j}+3\vec{k}$.
Find the angle between $\vec{a}$ and the plane consisting of $\vec{b}$ and $\vec{c}$.

$$\vec{b}\times\vec{c} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & -2 & 1 \\ 1 & 4 & 3 \end{vmatrix} = -10\vec{i}-8\vec{j}+14\vec{k}$$

$$\vec{a}\cdot(\vec{b}\times\vec{c}) = (2)(-10) + (3)(-8) + (-3)(14) = -86$$

$$|\vec{a}| = \sqrt{(2)^2+(3)^2+(-3)^2} = \sqrt{22}$$

$$|\vec{b}\times\vec{c}| = \sqrt{(-10)^2+(-8)^2+(14)^2} = 6\sqrt{10}$$

Angle between $\vec{a}$ and the plane consisting of $\vec{b}$ and $\vec{c}$

$$= \sin^{-1}\frac{|\vec{a}\cdot(\vec{b}\times\vec{c})|}{|\vec{a}||\vec{b}\times\vec{c}|}$$

$$= \sin^{-1}\frac{86}{(\sqrt{22})(6\sqrt{10})}$$

$$= \sin^{-1}\frac{43\sqrt{55}}{330} \qquad (= 75.1°, \text{ corr. to 3 sig. fig.})$$

Not for sale

Applications of Vector Product

[M2 2014 Q.8]

Let $\overrightarrow{OP} = -\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$, $\overrightarrow{OQ} = \mathbf{i} - \mathbf{j} + 2\mathbf{k}$ and $\overrightarrow{OR} = 2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$.

(a) Find $\overrightarrow{OP} \times \overrightarrow{OQ}$.

Hence find the volume of tetrahedron *OPQR*.

(b) Find the acute angle between the plane *OPQ* and the line *OR*, correct to the nearest $0.1°$.

(8 marks)

Not for sale

Applications of Vector Product

[M2 2016 Q.12 (b)]

Let $\overrightarrow{OA} = 2\mathbf{j} + 2\mathbf{k}$, $\overrightarrow{OB} = 4\mathbf{i} + \mathbf{j} + \mathbf{k}$, $\overrightarrow{OC} = 2\mathbf{i} - \mathbf{j} + 4\mathbf{k}$ and $\overrightarrow{OD} = 3\mathbf{i} + 2\mathbf{j} + 5\mathbf{k}$, where O is the origin. Denote the plane which contains A, B and C by Π.

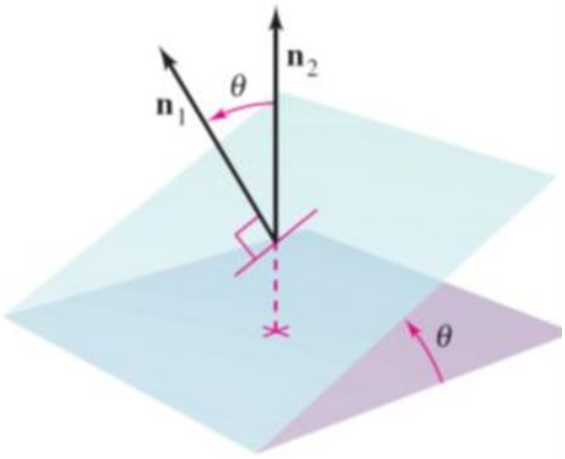
(i) Find a unit vector which is perpendicular to Π.

(ii) Find the angle between CD and Π. ***(6 marks)***

Not for sale

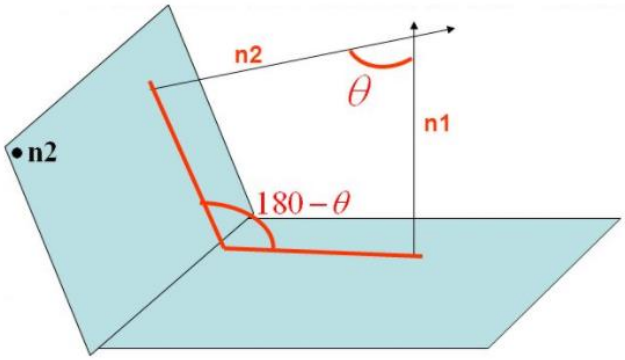
Application: Angle between two planes

Usually, the angle between two planes is equal to the angle between the two normal vectors of the planes:



$$\cos\theta = \frac{\overrightarrow{n_1} \bullet \overrightarrow{n_2}}{|\overrightarrow{n_1}||\overrightarrow{n_2}|}$$

But sometimes the angle between two planes maybe obtuse, and the angle between the two normal vectors θ $(= \cos^{-1}\frac{\overrightarrow{n_1} \bullet \overrightarrow{n_2}}{|\overrightarrow{n_1}||\overrightarrow{n_2}|})$ is acute, then the correct angle between two planes should be 180° – θ.



Not for sale

Applications of Vector Product

[M2 2018 Q.12 (c) Modified]

The position vectors of the points A, B, C and D are $4\mathbf{i} - 3\mathbf{j} + \mathbf{k}$, $-\mathbf{i} + 3\mathbf{j} - 3\mathbf{k}$, $7\mathbf{i} - \mathbf{j} + 5\mathbf{k}$ and $3\mathbf{i} - 2\mathbf{j} - 5\mathbf{k}$ respectively. Denote the plane which contains A, B and C by Π.

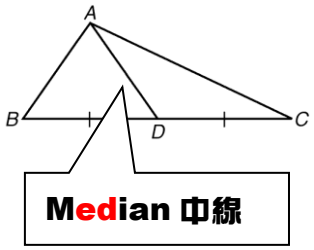
(i) Find $\overrightarrow{AB} \times \overrightarrow{AC}$ and $\overrightarrow{BC} \times \overrightarrow{BD}$

(ii) Find the angle between $\triangle BCD$ and Π. ***(5 marks)***

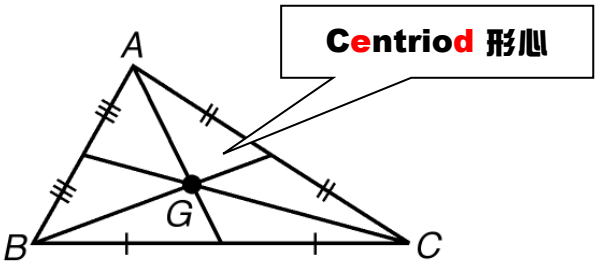
Not for sale

Bonus : Centers of Triangles

1.

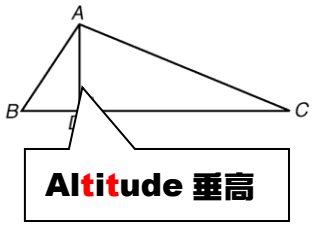


Median 中線

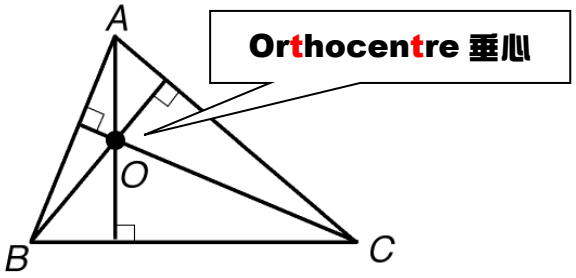


Centriod 形心

2.

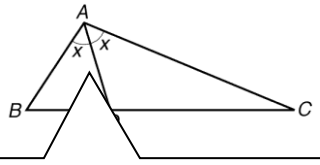


Altitude 垂高

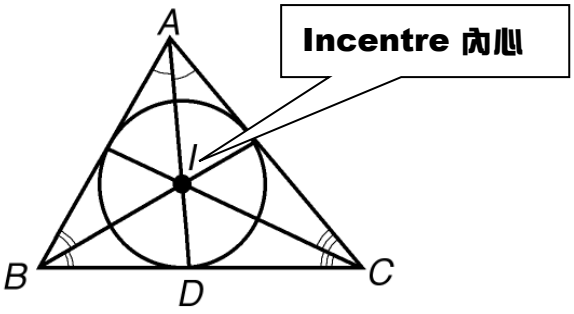


Orthocentre 垂心

3.

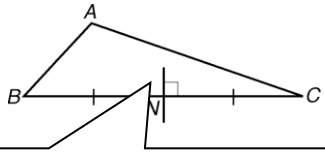


Angle bisector 角平分線

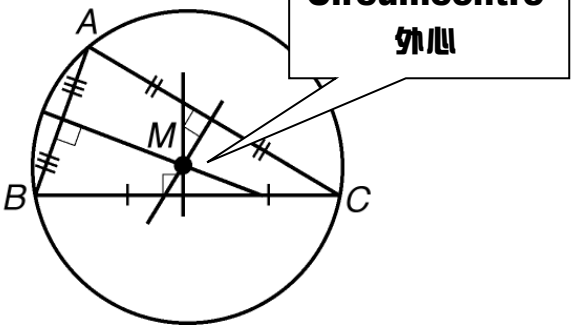


Incentre 內心

4.



Perpendicular bisector
垂直平分線



Circumcentre
外心

Not for sale

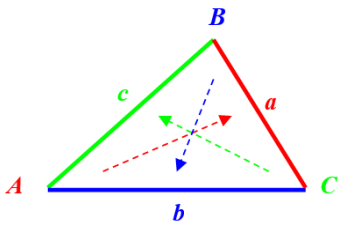
Bonus : Centers of Triangles

[M2 2022 Q.12(b) Modified]

Suppose that I is the incenter of $\triangle ABC$, then

$$\overrightarrow{OI} = \frac{a}{a+b+c}\overrightarrow{OA} + \frac{b}{a+b+c}\overrightarrow{OB} + \frac{c}{a+b+c}\overrightarrow{OC},$$

where $BC = a$, $AC = b$ and $AB = c$.



Given that $\overrightarrow{OA} = 35\mathbf{i} + 9\mathbf{j} + \mathbf{k}$, $\overrightarrow{OB} = 40\mathbf{i} - 3\mathbf{j} + \mathbf{k}$ and $\overrightarrow{OC} = -3\mathbf{j} + \mathbf{k}$.

(i) Find $\overrightarrow{OI}$.

(ii) By considering $\overrightarrow{AI} \times \overrightarrow{AB}$, find the radius of the inscribed circle of $\triangle ABC$.

(5 marks)

Not for sale

Bonus : Centers of Triangles

[M2 2017 Q.10(c) Modified]

ABC is a triangle. *D* is the mid-point of *AC*. *E* is a point on *BC*, such that *BE* : *EC* = 5 : 6. *AB* produced and *DE* produced meet at the point *F*. It is given that *DE* : *EF* = 1 :10. Let $\overrightarrow{OA}$ =2**i** + 3**j** – 2**k**, $\overrightarrow{OB}$ = 4**i** + 4**j** – **k**, $\overrightarrow{OC}$ = 8**i** – 3**j** – 2**k**, where *O* is the origin.

Let $\overrightarrow{OP}$ = 3**i** + 10**j** – 4**k**. Denote the circumcenter of $\triangle BCF$ by *Q*. Suppose *B*, *D*, *C* and *F* are concyclic, find the volume of the tetrahedron *ABPQ*. ***(3 marks)***

Not for sale

HKDSE Questions (level 3)

1. ***[M2 2015 Q.10]***

OAB is a triangle. *P* is the mid-point of *OA*. *Q* is a point lying on *AB* such that *AQ* : *QB* = 1 : 2 while *R* is a point lying on *OB* such that *OR* : *RB* = 3 : 1. *PR* and *OQ* intersect at *C*.

(a) (i) Let *t* be a constant such that $PC : CR = t : (1-t)$.
By expressing $\overrightarrow{OQ}$ in terms of $\overrightarrow{OA}$ and $\overrightarrow{OB}$, find the value of *t*.

(ii) Find *CQ* : *OQ*. ***(7 marks)***

(b) Suppose that $\overrightarrow{OA} = 20\mathbf{i} - 6\mathbf{j} - 12\mathbf{k}$, $\overrightarrow{OB} = 16\mathbf{i} - 16\mathbf{j}$ and $\overrightarrow{OD} = \mathbf{i} + 3\mathbf{j} - 6\mathbf{k}$, where *O* is the origin. Find

(i) the area of ΔOAB,

(ii) the volume of the tetrahedron *ABCD*. ***(5 marks)***

HKDSE Questions (level 3)

Not for sale

HKDSE Questions (Level 4)

2. ***[M2 2018 Q.12]***

The position vectors of the points *A*, *B*, *C* and *D* are $4\mathbf{i} - 3\mathbf{j} + \mathbf{k}$, $-\mathbf{i} + 3\mathbf{j} - 3\mathbf{k}$, $7\mathbf{i} - \mathbf{j} + 5\mathbf{k}$ and $3\mathbf{i} - 2\mathbf{j} - 5\mathbf{k}$ respectively. Denote the plane which contains *A*, *B* and *C* by Π. Let E be the projection of *D* on Π.

(a) Find

(i) $\overrightarrow{AB} \times \overrightarrow{AC}$,

(ii) the volume of the tetrahedron *ABCD*,

(iii) $\overrightarrow{DE}$. ***(5 marks)***

(b) Let *F* be a point lying on *BC* such that *DF* is perpendicular to *BC*.

(i) Find $\overrightarrow{DF}$.

(ii) Is $\overrightarrow{BC}$ perpendicular to $\overrightarrow{EF}$? Explain your answer. ***(5 marks)***

(c) Find the angle between $\triangle BCD$ and Π. ***(3 marks)***

HKDSE Questions (level 4)

Not for sale

HKDSE Questions (level 4)

3. ***[M2 2021 Q.12]***

The position vectors of the points A, B, C, D are $t\mathbf{i} + 14\mathbf{j} + s\mathbf{k}$, $12\mathbf{i} - s\mathbf{j} - 2\mathbf{k}$, $(s + 2)\mathbf{i} - 16\mathbf{j} + 10\mathbf{k}$ and $-t\mathbf{i} + (s + 2)\mathbf{j} + 14\mathbf{k}$ respectively, where $s, t \in \mathbf{R}$. Suppose that $\overrightarrow{AB}$ is parallel to $5\mathbf{i} - 4\mathbf{j} - 2\mathbf{k}$. Denote the plane which contains A, B and C by Π.

(a) Find

(i) s and t,

(ii) the area of $\triangle ABC$,

(iii) the volume of the tetrahedron $ABCD$,

(iv) the shortest distance from D to Π. ***(9 marks)***

(b) Let E be the projection of D on Π. Is E the circumcenter of $\triangle ABC$? Explain your answer. ***(4 marks)***

Not for sale

HKDSE Questions (level 4)

Not for sale

HKDSE Questions (level 3)

4. ***[M2 2019 Q.12]***

Let $\overrightarrow{OA} = \mathbf{i} - 4\mathbf{j} + 2\mathbf{k}$, $\overrightarrow{OB} = -5\mathbf{i} - 4\mathbf{j} + 8\mathbf{k}$ and $\overrightarrow{OC} = -5\mathbf{i} - 12\mathbf{j} + t\mathbf{k}$, where O is the origin and t is a constant. It is given that $\left|\overrightarrow{AC}\right| = \left|\overrightarrow{BC}\right|$.

(a) Find t. ***(3 marks)***

(b) Find $\overrightarrow{AB} \times \overrightarrow{AC}$. ***(2 marks)***

(c) Find the volume of the pyramid $OABC$. ***(2 marks)***

(d) Denote the plane which contains A, B and C be Π. It is given that P, Q and R are points lying on Π such that $\overrightarrow{OP} = p\mathbf{i}$, $\overrightarrow{OQ} = q\mathbf{j}$ and $\overrightarrow{OR} = r\mathbf{k}$. Let D be the projection of O on Π.

(i) Prove that $pqr \neq 0$.

(ii) Find $\overrightarrow{OD}$.

(iii) Let E be a point such that $\overrightarrow{OE} = \frac{1}{p}\mathbf{i} + \frac{1}{q}\mathbf{j} + \frac{1}{r}\mathbf{k}$. Describe the geometric relationship between D, E and O. Explain your answer.

(6 marks)

Not for sale

HKDSE Questions (level 3)

Not for sale

HKDSE Questions (Level 4)

5. ***[M2 2020 Q.12]***

Let $\overrightarrow{OP}$ = **i** + **j** + 4**k** and $\overrightarrow{OQ}$ = 5**i** – 7**j** – 4**k,** where O is the origin. R is a point lying on PQ such that $PR : RQ = 1 : 3$.

(a) Find $\overrightarrow{OP} \times \overrightarrow{OR}$. ***(2 marks)***

(b) Find $\overrightarrow{OS} = \overrightarrow{OP} + \overrightarrow{OR}$. Find the area of the quadrilateral $OPSR$. ***(2 marks)***

(c) Let N be a point such that $\overrightarrow{ON} = \lambda(\overrightarrow{OP} \times \overrightarrow{OR})$, where λ is a real number.

(i) Is $\overrightarrow{NR}$ perpendicular to $\overrightarrow{PQ}$? Explain your answer.

(ii) Let μ be a real number such that $\overrightarrow{NQ}$ is parallel to 11**i** + μ**j** – 10**k**.

(1) Find λ and μ.

(2) Denote the angle between $\triangle OPQ$ and $\triangle NPQ$ be θ.
Find $\tan\theta$. ***(8 marks)***

Not for sale

HKDSE Questions (level 4)

Not for sale

HKDSE Questions (Level 5)

6. ***[M2 Practice Paper Q.12]***

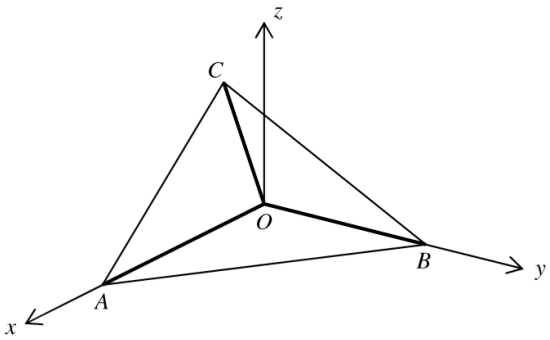


Figure 2

Let $\overrightarrow{OA}=\mathbf{i}$, $\overrightarrow{OB}=\mathbf{j}$ and $\overrightarrow{OC}=\mathbf{i}+\mathbf{j}+\mathbf{k}$ (see Figure 2). Let M and N be points on the straight lines AB and OC respectively such that $AM:MB=a:(1-a)$ and $ON:NC=b:(1-b)$, where $0<a<1$ and $0<b<1$. Suppose that MN is perpendicular to both AB and OC.

(a) (i) Show that $\overrightarrow{MN}=(a+b-1)\mathbf{i}+(b-a)\mathbf{j}+b\mathbf{k}$.

(ii) Find the values of a and b.

(iii) Find the shortest distance between the straight lines AB and OC.

(8 marks)

(b) (i) Find $\overrightarrow{AB}\times\overrightarrow{AC}$.

(ii) Let G be the projection of O on the plane ABC, find the coordinates of the intersecting point of the two straight lines OG and MN.

(5 marks)

Not for sale

HKDSE Questions (Level 5)

Not for sale

HKDSE Questions (Level ∞)

7. ***[M2 2013 Q.14]***

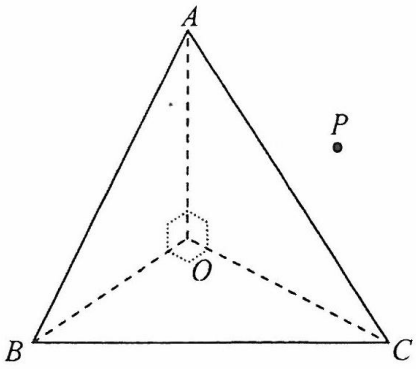


Figure 5

Figure 5 shows a fixed tetrahedron $OABC$ with $\angle AOB = \angle BOC = \angle COA = 90°$. P is a variable point such that $\overrightarrow{AP} \cdot \overrightarrow{BP} + \overrightarrow{BP} \cdot \overrightarrow{CP} + \overrightarrow{CP} \cdot \overrightarrow{AP} = 0$. Let D be the fixed point such that $\overrightarrow{OD} = \dfrac{\overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC}}{3}$. Let $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$, $\overrightarrow{OC} = \mathbf{c}$, $\overrightarrow{OP} = \mathbf{p}$ and $\overrightarrow{OD} = \mathbf{d}$.

(a) (i) Show that $\overrightarrow{AP} \cdot \overrightarrow{BP} = \mathbf{p} \cdot \mathbf{p} - (\mathbf{a} + \mathbf{b}) \cdot \mathbf{p}$.

(ii) Using (a)(i), show that $\mathbf{p} \cdot \mathbf{p} = 2\mathbf{p} \cdot \mathbf{d}$.

(iii) Show that $|\mathbf{p} - \mathbf{d}| = |\mathbf{d}|$. ***(8 marks)***

(b) (i) Alice claims that O lies on the sphere mentioned in (a)(iii). Do you agree? Explain your answer.

(ii) Suppose P_1, P_2, and P_3 are three distinct points on the sphere in (a)(iii) such that $\overrightarrow{DP_1} \times \overrightarrow{DP_2} = \overrightarrow{DP_2} \times \overrightarrow{DP_3}$. Alice claims that the radius of the circle passing through P_1, P_2, and P_3 is OD. Do you agree? Explain your answer. ***(4 marks)***

HKDSE Questions (level ∞)

Title of the publication:	Vector Product
Copyright owner:	HKEAA
Date of making this copy:	1 – 9 – 2023
Some material in this note has been copied under a license issued by the HKEAA. You are not permitted to make any further copy of this work, or to make It available to others. It is important to understand and respect copyright.	